

APMA 350 - Midterm II Cheat Sheet

$$1 + \tan^2 = \sec^2$$

$$\cos(-x) = \cos x$$

$$\sin(-x) = -\sin x$$

$$\cos 2x = \cos^2 x - \sin^2 x$$

$$\sin 2x = 2 \sin x \cos x$$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

Differential operators (factoring method) - 2nd order ODEs

$$Dy = y', D^2y = y'' \quad \therefore \text{e.g. } y'' - 5y' + 6y = 0 \Rightarrow D^2y - 5Dy + 6y = 0$$

$$\Rightarrow (D^2 - 5D + 6)y = 0 \Rightarrow (D-2)(D-3)y = 0$$

$$\text{let } (D-3)y = z \Rightarrow (D-2)z = 0 \Rightarrow Dz - 2z = 0 \Rightarrow z' - 2z = 0 \Rightarrow z' = 2z \Rightarrow z = Ae^{2t}$$

$$\text{But } z = (D-3)y \Rightarrow Ae^{2t} = Dy - 3y \Rightarrow y' - 3y = Ae^{2t} \quad (\text{use integrating factor})$$

Resonance, repeated roots: If 2nd order ODE has 1 root repeated twice, then $y = Ae^{rt} + Bte^{rt}$

$$\text{why? Consider } (e^{-2t}y)'' = (-2e^{-2t}y + e^{-2t}y')' = 4e^{-2t}y - 4e^{-2t}y' + e^{-2t}y''$$

$$\boxed{\begin{array}{l} \text{If root } r=2 \\ \therefore y = e^{2t}(uw) \\ \Rightarrow ye^{-2t} = (uw) \end{array}}$$

$$= e^{-2t}(y'' - 4y' + 4y) = e^{-2t}(0) = 0$$

original ODE = 0

$$\therefore (e^{-2t}y)'' = 0 \Rightarrow (e^{-2t}y)' = B \Rightarrow e^{-2t}y = Bt + A$$

$$\Rightarrow y = Ae^{-2t} + Bte^{-2t}$$

Complex roots: if $r = \pm i \Rightarrow y = A \cos t + B \sin t$. why? $y = Ae^{it} + Be^{-it}$

$$\Rightarrow y = A(\cos t + i \sin t) + B(\cos(-t) + i \sin(-t)) = A \cos t + iA \sin t + B \cos t - iB \sin t$$

$$= \underbrace{(A+B)}_A \cos t + \underbrace{i(A-B)}_B \sin t \Rightarrow y = A \cos t + B \sin t$$

$$\text{If } r = -3 \pm i \text{ (e.g.) } y = Ae^{-3t} \cos t + Be^{-3t} \sin t$$

Boundary-value problems: $y'' = \lambda y \quad \therefore \text{aux: } r^2 = \lambda \quad (\lambda > 0, \lambda = 0, \lambda < 0)$

$$\text{Case 1: } \lambda > 0, \lambda = \omega^2 \quad (\omega > 0) \Rightarrow r^2 = \omega^2 \Rightarrow r = \pm \omega \Rightarrow y = Ae^{\omega t} + Be^{-\omega t}$$

$$y(0) = 0 \Rightarrow 0 = Ae^0 + Be^0 \Rightarrow A + B = 0 \Rightarrow B = -A \Rightarrow y = Ae^{\omega t} - Ae^{-\omega t}$$

$$y(\pi) = 0 \Rightarrow 0 = Ae^{\omega \pi} - Ae^{-\omega \pi} \Rightarrow Ae^{\omega \pi} = Ae^{-\omega \pi} \Rightarrow \omega \pi = -\omega \pi \Rightarrow \omega = -\omega \Rightarrow \omega = 0$$

(contradiction as λ must be > 0)

$$\text{Case 2: } \lambda = 0 \Rightarrow r^2 = 0 \Rightarrow r = 0 \text{ (repeated twice)} \Rightarrow y = Ae^{0t} + Bte^{0t} = A + Bt$$

$$y(0) = 0 \Rightarrow 0 = A + B(0) \Rightarrow A = 0 \quad \therefore y = Bt, \quad y(\pi) = 0 \Rightarrow 0 = B\pi \Rightarrow B = 0, \text{ but then } y = 0$$

$$\text{Case 3: } \lambda < 0, \lambda = -(\omega^2) \quad (\omega > 0) \Rightarrow r^2 = -\omega^2 \Rightarrow r = \pm i\omega \quad (\text{contradiction as we want non zero solns.})$$

$$\therefore y = A \cos \omega t + B \sin \omega t, \quad y(0) = 0 \Rightarrow 0 = A \cos 0 + B \sin 0 \Rightarrow A = 0$$

$$\Rightarrow y = B \sin \omega t, \quad y(\pi) = 0 \Rightarrow 0 = B \sin \omega \pi \Rightarrow \sin \omega \pi = 0 \Rightarrow \omega \pi = \pi m \Rightarrow \omega = m$$

$$\therefore \text{Eigenvalues: } \lambda = -m^2 \quad (m = 1, 2, \dots) \quad \left. \begin{array}{l} \omega > 0 \\ \Rightarrow \lambda = -\omega^2 \\ = -m^2 \end{array} \right\}$$

$$\text{Eigenfunctions: } y = \sin mt \quad (m = 1, 2, \dots) \quad \therefore \omega = m = 1, 2, \dots$$

undetermined coefficients: $y = y_0 + y_p$ (1st solve homogeneous eqn., then find particular soln)

$$\text{e.g. } y_p \text{ of } y'' - 5y' + 6y = e^t$$

$$\text{let } y_p = Ae^t \rightarrow \text{plug } y_p \text{ into ODE} \Rightarrow A = \frac{1}{2} \quad \therefore y_p = \frac{1}{2} e^t \quad \therefore y = y_0 + \frac{1}{2} e^t$$

$$\text{Rules: } \bullet \text{ If RHS is } e^{rt}, y_p = Ae^{rt} \quad \bullet \text{ If RHS is polynomial } (t^2), y_p = At^2 + Bt + C$$

$$\bullet \text{ If RHS is } \frac{\sin t}{\cos t}, y_p = A \sin t + B \cos t \quad \bullet \text{ If RHS is } f(t) + g(t), \text{ Find } y_p \text{ for } f(t)$$

$$\therefore y_p = y_{p1} + y_{p2} \quad y_{p2} \text{ for } g(t)$$

Resonance in undetermined coeffs

If the RHS $e^{rt} \rightarrow r = r_1$, corresponds with a root $r = r_1$ of the homogeneous,

$$\text{then } y_p = Ate^{r_1 t} \quad \text{e.g. } y'' - 5y' + 6y = e^{2t}$$

$$\begin{array}{c} r=2, 3 \\ \uparrow \quad \uparrow \end{array}$$

$$\therefore y_p = Ate^{2t}$$

$$y_0 = Ae^{2t} + Ae^{3t}$$

ALWAYS CHECK FOR RESONANCE !!! especially when asked "what's the form

$$\text{e.g. RHS} = t^3 e^{-4t} \rightarrow r = -4$$

$$y_p = (At^3 + Bt^2 + Ct + D)e^{-4t}$$

$$\text{Resonance, } y_p = t(At^2 + Bt^2 + Ct + D)e^{-4t}$$

$$\text{RHS} = e^{-4t} \cos t \rightarrow r = -4 \pm i$$

$$\therefore y_p = Ae^{-4t} \cos t + Be^{-4t} \sin t$$

$$\text{RHS} = t = te^{0t} \rightarrow r = 0 \text{ of } y_p$$

$$\text{If resonance, } y_p = t(A + B)$$

$$\text{If RHS root resonates w/ repeated root in } y_0, y_p = At^2 e^{rt}$$

→ Var of par: $\begin{bmatrix} -5 & 3 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 9 \\ -5 \end{bmatrix}$; $x = \frac{\begin{vmatrix} 9 & 3 \\ -5 & -1 \end{vmatrix}}{\begin{vmatrix} -5 & 3 \\ 3 & -1 \end{vmatrix}} = \frac{(9)(-1) - (3)(-5)}{(-5)(-1) - (3)(3)} = \frac{-9 + 15}{5 - 9} = \frac{6}{-4} = -\frac{3}{2}$

e.g. $y'' + y = \sec t$

1) Put into standard form (coeff of y'' is 1)

2) Homogeneous; $y_0 = A \cos t + B \sin t$

3) y_p is of same form as $y_0 \therefore y_p = u(t) \cos t + v(t) \sin t$

4) $y_p' = \cos t u'(t) + \sin t v'(t) - u(t) \sin t + v(t) \cos t \therefore u'(t) \cos t + v'(t) \sin t = 0$ (1)

$\therefore y_p' = -u(t) \sin t + v(t) \cos t \Rightarrow y_p'' = -u'(t) \sin t - u(t) \cos t + v'(t) \cos t - v(t) \sin t$

5) $y_p'' + y_p$ (original ODE) = (after adding and cancelling terms) = $-u'(t) \sin t + v'(t) \cos t = \sec t$ (RHS)

$\therefore$ From (1) and (2), $\begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix} \begin{bmatrix} u'(t) \\ v'(t) \end{bmatrix} = \begin{bmatrix} 0 \\ \sec t \end{bmatrix}$

6) Use Cramer's rule to find $u'(t)$, $v'(t)$, integrate to find $u(t)$, $v(t)$

→ Laplace: $\mathcal{L}\{f(t)\} = \int_0^\infty f(t) e^{-st} dt$, $\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0)$, $\mathcal{L}\{t^n f(t)\} = (-1)^n \mathcal{L}\{f^{(n)}(t)\}$

Tabular integration:

e.g. $\int_0^\infty t^3 e^{-st} dt$

$+ t^3 \rightarrow e^{-st} / -s$

$+ 6t^2 \rightarrow e^{-st} / (-s)^2$

$+ 6t \rightarrow e^{-st} / (-s)^3$

$+ 0 \rightarrow e^{-st} / (-s)^4$

$+ \dots \rightarrow \dots$

Terms at ∞ are 0

[compare real and imaginary parts to find] and $= \frac{1}{s - ai} = \frac{s + ai}{s^2 + a^2} = \frac{s}{s^2 + a^2} + \frac{ai}{s^2 + a^2}$

→ Laplace to solve ODEs: Take Laplace of both sides

get $\mathcal{L}\{y\}$ on one side, something else on the other. Use partial fractions to solve. (or completing the square)

Partial fractions:

$\frac{f(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$ | $\frac{f(x)}{(x-a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}$ | $\frac{f(x)}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$

Completing the square: e.g. $x^2 + 4x + 7 = (x + \frac{4}{2})^2 + A = (x+2)^2 + A$
 $= x^2 + 4x + 4 + A \Rightarrow A = 3$

→ Step functions: e.g. $f(t) = \begin{cases} 2 & 0 \leq t < 4 \\ 5 & 4 \leq t < 7 \\ 0 & t \geq 7 \end{cases} = 2 + (5-2)u_4 = 2 + 3u_4(t)$

$\mathcal{L}\{f(t-c)u_c(t)\} = e^{-cs} \mathcal{L}\{f(t)\}$ (shift jump)

$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$ shift where $F(s)$ is $\mathcal{L}\{f(t)\}$

(Here complete the square to find shifted (replace in ODE))

→ Dirac delta: $\delta(t)$: 1) $\delta(t) = 0$, except at $t=0$, $\delta(t) = \infty$, 2) $\int_{-\infty}^\infty \delta(t) dt = 1$

$\delta(t) = \lim_{n \rightarrow \infty} n f(nt)$. $\mathcal{L}\{\delta(t)\} = ? \rightarrow \int_{-\infty}^\infty \delta(t) f(t) dt = f(0) \therefore \mathcal{L}\{\delta(t)\} = \int_0^\infty \delta(t) e^{-st} dt = e^{-s(0)} = 1$

→ Convolution: $\mathcal{L}\{f(t)g(t)\} \neq \mathcal{L}\{f(t)\}\mathcal{L}\{g(t)\}$

But $(f * g)(t) = \int_0^t f(t-\tau) g(\tau) d\tau = \int_0^t f(\tau) g(t-\tau) d\tau$ and $\mathcal{L}\{(f * g)(t)\} = \mathcal{L}\{f(t)\}\mathcal{L}\{g(t)\}$

Use in ODEs to find y from $\mathcal{L}\{y\}$ e.g. $y'' - 4y' + 4y = \tan t$, take Laplace: $\mathcal{L}\{y\} + \mathcal{L}\{t * y\} = \frac{8}{3(s^2+9)}$

→ Integral equations:

$\phi(t) + \int_0^t (t-\tau)\phi(\tau) d\tau = \frac{8}{3} \sin 3t$
 $(t * \phi) \Rightarrow \phi + (t * \phi) = \frac{8}{3} \sin 3t$
 $\Rightarrow \mathcal{L}\{\phi\} + \mathcal{L}\{t * \phi\} = \frac{8}{3} \mathcal{L}\{\sin 3t\}$
 $\Rightarrow \mathcal{L}\{\phi\} (1 + \frac{1}{s^2}) = \frac{8}{3} \frac{3}{s^2+9}$
 $\Rightarrow \mathcal{L}\{\phi\} = \frac{8s^2}{(s^2+1)(s^2+9)}$